

Generation Interconnection System Impact Study Report

For Queue Project Q22-2157

Alvin 138 kV
November 17, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2157. The Interconnection Customer proposes a 94.6 MW solar plus storage facility interconnecting at the Alvin 138 kV substation in the Virginia area. The request was received on August 14, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 652 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,659,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Springfield Power Llc / Edinburg 138kV, the Buckhorn / Templeton 138kV loads to 130.2% of its 172 MVA rating. Project DFAX: 9.2%.

For the loss of Springfield Power Llc Substation / Edinburg 138kV, the Roslyn / Buckhorn 138kV loads to 127.8% of its 177 MVA rating. Project DFAX: 9.2%.

For the loss of Edinburg / Springfield Power Llc 138kV, the Templeton / Red Tail 138kV Line loads to 119.5% of its 186 MVA rating. Project DFAX: 9.2%.

For the loss of Purdy / Chatom 138kV, the Timberville / Frito Lay 138kV L loads to 107.3% of its 322 MVA rating. Project DFAX: 21.2%.

New Salisbury / Timberville 138kV L: 107.2% loading (328 MVA) following the outage of the Chatom / Purdy 138kV Line; DFAX = 21.2%.

The Plant City / Red Tail 138kV is overloaded to 105.2% of its 203 MVA emergency rating for the loss of the Edinburg / Springfield Power Llc 138kV. The project contributes 9.2% DFAX (8.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Buckhorn / Templeton 138kV	Rebuild line with higher-capacity conductor	\$5,515,000	5% (\$277,000)
Roslyn / Buckhorn 138kV	Rebuild line with higher-capacity conductor	\$8,415,000	5% (\$422,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Templeton / Red Tail 138kV Line	Rebuild line with higher-capacity conductor	\$12,516,000	5% (\$628,000)
Timberville / Frito Lay 138kV	Reconductor line	\$1,127,000	6% (\$74,000)
New Salisbury / Timberville 138kV L	Reconductor line	\$2,485,000	6% (\$162,000)
Plant City / Red Tail 138kV Line	Reconductor line	\$1,915,000	5% (\$96,000)
	Total allocated to project		\$1,659,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.